

ASSIGNMENTN – 2

DATA EXPLORATION – TASK EXECUTION - SCREENSHOTS

Question :-1. Handling Missing Values:

1. Check for missing values in the 'Price' column. How would you handle products with missing price information?

	A	B	C	D	E	F	G
	Product Name	Brand Name		Price (\$)	Quantity	Category	
28-JAN-US	Laptop	Dell	1000	1000	30	Electronics	
15-FEB-US	Sneakers	Nike	80	80	15	Fashion	
03-MAR-US	Coffee Maker	Keurig	130	130	40	Kitchen	
11-APR-US	Smartphone	Samsung	900	900	25	Electronics	
22-MAY-US	Backpack	North Face	70	70	20		
07-JUN-UK	Headphones	Sony	309.3548387		45	Electronics	

- Missing price values were replaced with the average of the available prices in the dataset. This method was used to preserve all records and provide reasonable estimates for missing values, ensuring consistency in subsequent analysis.

2. If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.

=COUNTBLANK(\$F\$2:\$F\$35)						
A	B	C	D	E	F	G
Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	
28-JAN-US	Laptop	Dell	1000	30	Electronics	4
15-FEB-US	Sneakers	Nike	80	15	Fashion	
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	
11-APR-US	Smartphone	Samsung	900	25	Electronics	
22-MAY-US	Backpack	North Face	70	20		
07-JUN-UK	Headphones	Sony	309.35	45	Electronics	
19-JUL-UK	T-Shirt	Adidas	30	5	Fashion	
23-AUG-UK	Blender	Ninja	90	35	Kitchen	

Clipboard

Find and Replace

Find what: Backpack

Options >>

Find All Find Next Close

Book Sheet Nam

Excel Assignment 2 - Data Cleaning and Transformation data.xlsx Dataset

1 cell(s) found

Product ID	Product Name	Brand	Price	Category
28-JAN-US	Laptop	Dell	500	Electronics
15-FEB-US	Sneakers	Nike	50	Electronics
03-MAR-US	Coffee Maker	Keurig	50	Kitchen
11-APR-US	Smartphone	Samsung	50	Electronics
22-MAY-US	Backpack	Northern	50	Electronics
07-JUN-UK	Headphones	Sony	50	Electronics
19-JUL-UK	T-Shirt	Adidas	50	Electronics
23-AUG-UK	Blender	Ninja	50	Kitchen
05-SEP-UK	Tablet	Apple	500	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10 Outdoor
17-JUN-IN	Laptop	HP	950	25 Electronics
25-NOV-AU	Sneakers	Adidas	90	40

Format Painter

Clipboard

Find and Replace

Find what: Sneakers

Options >>

Find All Find Next Close

Sheet	Name	Cell	Value	Formula
and Transformation data.xlsx	Dataset	\$B\$3	Sneakers	
and Transformation data.xlsx	Dataset	\$B\$13	Sneakers	

2 cell(s) found

Product ID	Product Name	Brand	Price	Category
JAN-US	Laptop	Dell	500	Electronics
FEB-US	Sneakers	Nike	50	Electronics
MAR-US	Coffee Maker	Keurig	50	Kitchen
APR-US	Smartphone	Samsung	50	Electronics
MAY-US	Backpack	Northern	50	Electronics
JUN-UK	Headphones	Sony	50	Electronics
JUL-UK	T-Shirt	Adidas	50	Electronics
AUG-UK	Blender	Ninja	50	Kitchen
SEP-UK	Tablet	Apple	500	Electronics
OCT-UK	Hiking Boots	Timberland	130	10 Outdoor
JUN-IN	Laptop	HP	950	25 Electronics
NOV-AU	Sneakers	Adidas	90	40
DEC-DE	Coffee Maker	Nespresso	120	35
FEB-CA	Smartwatch	Fitbit	150	15 Electroni
APR-ES	Headphones	Bose	250	20 Electroni

Format Painter

Clipboard

Find and Replace

Find what: Coffee Maker

Options >>

Find All Find Next Close

Sheet	Name	Cell	Value	Formula
information data.xlsx	Dataset	\$B\$4	Coffee Maker	
information data.xlsx	Dataset	\$B\$14	Coffee Maker	

2 cell(s) found

Clipboard

Font

Alignment

Find and Replace

Find what: Fitness Tracker

Options >>

Find All Find Next Close

Sheet	Name	Cell	Value	Formula
information data.xlsx	Dataset	\$B\$24	Fitness Tracker	
information data.xlsx	Dataset	\$B\$32	Fitness Tracker	

2 cell(s) found

Product ID	Product Name	Category
8-JAN-US	Laptop	Electronics
5-FEB-US	Sneakers	Footwear
3-MAR-US	Coffee Maker	Kitchen
1-APR-US	Smartphone	Electronics
2-MAY-US	Backpack	Outdoor
7-JUN-UK	Headphones	Electronics
9-JUL-UK	T-Shirt	Apparel
3-AUG-UK	Blender	Kitchen
5-SEP-UK	Tablet	Electronics
4-OCT-UK	Hiking Boots	Outdoor

Product ID	Product Name	Category
6-NOV-AU	Sneakers	Footwear
8-DEC-DE	Coffee Maker	Kitchen
3-FEB-CA	Smartwatch	Electronics
6-APR-ES	Headphones	Electronics
1-AUG-CA	Laptop	Electronics
9-AUG-CN	Smartwatch	Electronics
7-JAN-IT	Laptop	Electronics
1-MAR-UK	Sunglasses	Fashion
1-AUG-US	Camping	Outdoor
1-MAY-RU	Camera	Electronics
9-JAN-CA	Microwave	Kitchen
9-JUL-BR	Fitness Tracker	Electronics
1-AUG-CA	Laptop Bag	Accessories
9-SEP-CA	Smartphone	Electronics
3-JUN-CA	Sunglasses	Fashion
1-JUL-CA	Blender	Kitchen
5-APR-ES	Headphones	Electronics

COUNTBLANK was used to identify 4 missing category values. Using the Find (Ctrl + F) feature, matching product records were searched within the dataset. Three products had existing category values and were filled accordingly. Since Backpack had no matching record, it was categorized manually as Outdoor based on the product type.

Question :- 2. Correcting Inconsistent Data:

1. Identify any inconsistent text formats present in the "Product Name" column.

Formula Bar: `=PROPER(C2)`

A	B	C	D	E
	Product Name	Brand Name	Price (\$)	
28-JAN-US	Laptop	laptop	Dell	1000
15-FEB-US	Sneakers	Sneakers	Nike	80
03-MAR-US	Coffee Maker	Coffee Maker	Keurig	130
11-APR-US	Smartphone	smartphone	Samsung	900
22-MAY-US	Backpack	Backpack	North Face	70
07-JUN-UK	Headphones	Headphones	Sony	309.35
19-JUL-UK	T-Shirt	T-shirt	Adidas	30

Formula Bar: `=TRIM(CLEAN(C2))`

A	B	C	D	E	F
	Product Name	Brand Name	Price (\$)	Quantity	
28-JAN-US	Laptop	Laptop	Dell	1000	
15-FEB-US	Sneakers	Sneakers	Nike	80	
03-MAR-US	Coffee Maker	Coffee Maker	Keurig	130	
11-APR-US	Smartphone	Smartphone	Samsung	900	
22-MAY-US	Backpack	Backpack	North Face	70	
07-JUN-UK	Headphones	Headphones	Sony	309.35	
19-JUL-UK	T-Shirt	T-Shirt	Adidas	30	
23-AUG-UK	Blender	Blender	Ninja	90	
05-SEP-UK	Tablet	Tablet	Apple	500	
14-OCT-UK	Hiking Boots	Hiking Boots	Timberland	130	
17-JUN-IN	Laptop	Laptop	HP	950	
25-NOV-AU	Sneakers	Sneakers	Adidas	90	
08-DEC-DE	Coffee Maker	Coffee Maker	Nespresso	120	
18-FEB-CA	Smartwatch	Smartwatch	Fitbit	150	

Formula Bar: `=TRIM(CLEAN(D2))`

A	B	C	D	E	F
	Product Name	Brand Name	Price (\$)	Quantity	
28-JAN-US	Laptop	Dell	Dell	1000	
15-FEB-US	Sneakers	Nike	Nike	80	
03-MAR-US	Coffee Maker	Keurig	Keurig	130	
11-APR-US	Smartphone	Samsung	Samsung	900	
22-MAY-US	Backpack	North Face	North Face	70	
07-JUN-UK	Headphones	Sony	Sony	309.35	
19-JUL-UK	T-Shirt	Adidas	Adidas	30	
23-AUG-UK	Blender	Ninja	Ninja	90	

- Inconsistent text formats in the Product Name column were identified and standardized using the PROPER, TRIM and CLEAN functions to ensure a consistent naming format throughout the dataset.

- Identify any typos present in the "Category" column.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	120	10	Kitchen
11-APR-US	Smartphone	Samsung	600	20	Electronics
22-MAY-US	Backpack	North Face	50	35	Accessories
07-JUN-UK	Headphones	Sony	150	15	Electronics
19-JUL-UK	T-Shirt	Adidas	25	40	Fashion
23-AUG-UK	Blender	Ninja	90	12	Kitchen
05-SEP-UK	Tablet	Apple	400	18	Electronics
14-OCT-UK	Hiking Boots	Timberland	110	10	Fashion
17-JUN-IN	Laptop	HP	750	25	Electronics
25-NOV-AU	Sneakers	Adidas	70	20	Fashion
08-DEC-DE	Coffee Maker	Nespresso	130	10	Kitchen
18-FEB-CA	Smartwatch	Fitbit	180	12	Electronics
16-APR-ES	Headphones	Bose	200	10	Electronics
21-AUG-CA	Laptop Bag	Samsonite	60	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	100	10	Outdoor

- The Category column was reviewed using filters to identify any spelling mistakes or typographical errors. A typo was found in the "Electronics" category.

- Use the find and replace function to standardize the text formats in the "Product Name" column and fix any typos or misspellings in the "Category" column.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	120	10	Kitchen
11-APR-US	Smartphone	Samsung	600	20	Electronics
22-MAY-US	Backpack	North Face	50	35	Accessories
07-JUN-UK	Headphones	Sony	150	15	Electronics
19-JUL-UK	T-Shirt	Adidas	25	40	Fashion
23-AUG-UK	Blender	Ninja	90	12	Kitchen
05-SEP-UK	Tablet	Apple	400	18	Electronics
14-OCT-UK	Hiking Boots	Timberland	110	10	Fashion
17-JUN-IN	Laptop	HP	750	25	Electronics
25-NOV-AU	Sneakers	Adidas	70	20	Fashion
08-DEC-DE	Coffee Maker	Nespresso	130	10	Kitchen
18-FEB-CA	Smartwatch	Fitbit	180	12	Electronics
16-APR-ES	Headphones	Bose	200	10	Electronics
21-AUG-CA	Laptop Bag	Samsonite	60	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	100	10	Outdoor

- Applied Find and Replace to detect and fix formatting inconsistencies in the Product Name column. This ensured a uniform and standardized naming format throughout the dataset.

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
8-JAN-US	Laptop	Dell	1000	30	Electronics
5-FEB-US	Smartwatch	Apple	250	15	Electronics
3-MAR-US	Camera	Canon	450	10	Electronics
1-APR-US	Smartphone	Samsung	600	20	Electronics
2-MAY-US	Backpack	Patagonia	120	5	Accessories
7-JUN-UK	Hiking Boots	Merrell	180	10	Footwear
9-JUL-UK	T-Shirt	Adidas	25	50	Clothing
3-AUG-UK	Bleach	Clorox	15	10	Household
5-SEP-UK	Tatooine	Star Wars	10	5	Collectibles
4-OCT-UK	Hiking	Osprey	150	10	Accessories
7-JUN-IN	Laptop	ASUS	980	10	Electronics
5-NOV-AU	Smartwatch	Apple	250	15	Electronics
8-DEC-DE	Smartwatch	Apple	250	15	Electronics
8-FEB-CA	Smartwatch	Apple	250	15	Electronics
6-APR-ES	Smartwatch	Apple	250	15	Electronics
1-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
0-AUG-CN	Smartwatch	Huawei	160	15	Electronics
7-JAN-IT	Laptop	Asus	980	10	Electronics
1-MAR-UK	Sunglasses	Oakley	150	15	Fashion
4-AUG-US	Camping Tent	Coleman	309.35	10	Outdoor
4-MAY-RU	Camera	Nikon	700	50	Electronics
8-JAN-CA	Smartwatch	Apple	250	15	Electronics

- The Category column was reviewed using filters to identify inconsistent spellings and typographical errors. A typo was found in the "Electronics" category and was corrected using the Find and Replace feature.

3. Removing Duplicates:

1. Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.

	A	B	C	D	E	F
12	17-JUN-IN	Laptop	HP	950	25	Electronics
13	25-NOV-AU	Sneakers	Adidas	90	40	
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16	16-APR-ES	Headphones	Bose	250	20	Electronics
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
19	27-JAN-IT	Laptop	Asus	980	10	Electronics
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
21	14-AUG-US	Camping Tent	Coleman	309.35	10	Outdoor
22	14-MAY-RU	Camera	Nikon	700	50	Electronics
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
26	29-SEP-CA	Smartphone	Google	800	45	Electronics
27	03-JUN-CA	Sunglasses	Ray-Ban	309.35	25	Fashion
28	11-JUL-CA	Blender	Vitamix	400	40	Kitchen
29	16-APR-ES	Headphones	Bose	250	20	Electronics
30	07-MAR-CA	Dress	Zara	60	30	Fashion
31	13-APR-CA	Toaster	Hamilton	40	10	Kitchen
32	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
33	02-DEC-CA	Jeans	Levi's	50	50	Fashion
34	17-JUN-IN	Laptop	HP	950	25	Electronics
35	09-JUL-ES	Watch	Casio	100	20	Accessories

- As an initial data cleaning step, the dataset was checked for duplicate records using the Conditional Formatting → Highlight Cells Rules → Duplicate Values feature. Duplicate records were identified and highlighted for review. However, the duplicate entries were retained because they may represent valid repeated purchases, and therefore no records were removed from the dataset.

4. Splitting and Merging Data:

1. Split the "Product ID" column into two separate columns for "Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.

=TRIM(CLEAN(B2))						
A	B	C	D	E	F	G
	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	11-APR-US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	22-MAY-US	Backpack	North Face	70	20	Outdoor
07-JUN-UK	07-JUN-UK	Headphones	Sony	309.35	45	Electronics
19-JUL-UK	19-JUL-UK	T-Shirt	Adidas	30	5	Fashion
23-AUG-UK	23-AUG-UK	Blender	Ninja	90	35	Kitchen

=LEFT(A2,6)							
A	B	C	D	E	F	G	H
Product ID	Manufacturing Date	Country Code	Brand Name	Price (\$)	Quantity	Category	
28-JAN-US	28-JAN	US	Laptop	Dell	₹ 1,000.00	30	Electronics
15-FEB-US	15-FEB	US	Sneakers	Nike	₹ 80.00	15	Fashion
03-MAR-US	03-MAR	US	Coffee Maker	Keurig	₹ 130.00	40	Kitchen
11-APR-US	11-APR	US	Smartphone	Samsung	₹ 900.00	25	Electronics
22-MAY-US	22-MAY	US	Backpack	North Face	₹ 70.00	20	Outdoor
07-JUN-UK	07-JUN	UK	Headphones	Sony	₹ 309.35	45	Electronics
19-JUL-UK	19-JUL	UK	T-Shirt	Adidas	₹ 30.00	5	Fashion
23-AUG-UK	23-AUG	UK	Blender	Ninja	₹ 90.00	35	Kitchen
05-SEP-UK	05-SEP	UK	Tablet	Apple	₹ 500.00	50	Electronics
14-OCT-UK	14-OCT	UK	Hiking Boots	Timberland	₹ 130.00	10	Outdoor
17-JUN-IN	17-JUN	IN	Laptop	Hp	₹ 950.00	25	Electronics

=RIGHT(A2,2)								
A	B	C	D	E	F	G	H	I
Product ID	Manufacturing Date	Country Code	Brand Name	Price (\$)	Quantity	Category	Product Brand	
28-JAN-US	28-JAN	US	Laptop	Dell	₹ 1,000.00	30	Electronics	Laptop Dell
15-FEB-US	15-FEB	US	Sneakers	Nike	₹ 80.00	15	Fashion	Sneakers Nike
03-MAR-US	03-MAR	US	Coffee Maker	Keurig	₹ 130.00	40	Kitchen	Coffee Maker Keurig
11-APR-US	11-APR	US	Smartphone	Samsung	₹ 900.00	25	Electronics	Smartphone Samsung
22-MAY-US	22-MAY	US	Backpack	North Face	₹ 70.00	20	Outdoor	Backpack North Face
07-JUN-UK	07-JUN	UK	Headphones	Sony	₹ 309.35	45	Electronics	Headphones Sony
19-JUL-UK	19-JUL	UK	T-Shirt	Adidas	₹ 30.00	5	Fashion	T-Shirt Adidas
23-AUG-UK	23-AUG	UK	Blender	Ninja	₹ 90.00	35	Kitchen	Blender Ninja
05-SEP-UK	05-SEP	UK	Tablet	Apple	₹ 500.00	50	Electronics	Tablet Apple
14-OCT-UK	14-OCT	UK	Hiking Boots	Timberland	₹ 130.00	10	Outdoor	Hiking Boots Timberland
17-JUN-IN	17-JUN	IN	Laptop	Hp	₹ 950.00	25	Electronics	Laptop Hp

- The Product ID column was divided into two separate columns for Manufacturing Date and Country Code. The date portion and country code were extracted from the Product ID using text functions, and the data was cleaned to retain only the required information.

2. Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".

Formula Bar: =D2&" "&E2

Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Price (\$)	Quantity	Category	Product Brand
JN-US	28-JAN	US	Laptop	Dell	₹ 1,000.00	30	Electronics	Laptop Dell
EB-US	15-FEB	US	Sneakers	Nike	₹ 80.00	15	Fashion	Sneakers Nike
AR-US	03-MAR	US	Coffee Maker	Keurig	₹ 130.00	40	Kitchen	Coffee Maker Keurig
PR-US	11-APR	US	Smartphone	Samsung	₹ 900.00	25	Electronics	Smartphone Samsung
AY-US	22-MAY	US	Backpack	North Face	₹ 70.00	20	Outdoor	Backpack North Face
IN-UK	07-JUN	UK	Headphones	Sony	₹ 309.35	45	Electronics	Headphones Sony
IL-UK	19-JUL	UK	T-Shirt	Adidas	₹ 30.00	5	Fashion	T-Shirt Adidas
JG-UK	23-AUG	UK	Blender	Ninja	₹ 90.00	35	Kitchen	Blender Ninja
EP-UK	05-SEP	UK	Tablet	Apple	₹ 500.00	50	Electronics	Tablet Apple

- The Product Name and Brand Name columns were combined into a new Product Brand column using the "&" operator. This created a single field containing both the product and brand information for easier analysis.

5. Number Formatting:

1. Format the data type of the "Price" column to a currency format.

Clipboard: Cut, Copy, Format Painter

Font: Arial, 10, Bold, Italic, Underline, Text Color, Background Color

Alignment: Left, Center, Right, Justify, Merge & Center

Number: Currency, \$, %, ' (Thousands Separator)

Formula Bar: 1000

Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Product Brand
8-JAN-US	Laptop	Dell	₹ 1,000.00	30	Electronics	Laptop Dell
5-FEB-US	Sneakers	Nike	₹ 80.00	15	Fashion	Sneakers Nike
3-MAR-US	Coffee Maker	Keurig	₹ 130.00	40	Kitchen	Coffee Maker Keurig
1-APR-US	Smartphone	Samsung	₹ 900.00	25	Electronics	Smartphone Samsung
2-MAY-US	Backpack	North Face	₹ 70.00	20	Outdoor	Backpack North Face
7-JUN-UK	Headphones	Sony	₹ 309.35	45	Electronics	Headphones Sony
9-JUL-UK	T-Shirt	Adidas	₹ 30.00	5	Fashion	T-Shirt Adidas
3-AUG-UK	Blender	Ninja	₹ 90.00	35	Kitchen	Blender Ninja

- The Price column was converted to Currency format to display product prices in a standardized monetary format.

2.Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY " format.

Cut

Copy

Format Painter

Clipboard

Arial10A A

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Font

Alignment

Wrap Text

Merge & Center

Alignment

Date

\$ % ' .00 .00

Number

Conditional Formatting

Formulas

Styles

:

X

✓

f x

=LEFT(A2,6)

A	B	C	D	E	F	G	H	I
Product ID	Manufacturing Date	Country Code		Brand Name	Price (\$)	Quantity	Category	Product Brand
3-JAN-US	28-JAN	US	Laptop	Dell	₹ 1,000.00	30	Electronics	Laptop Dell
5-FEB-US	15-FEB	US	Sneakers	Nike	₹ 80.00	15	Fashion	Sneakers Nike
3-MAR-US	03-MAR	US	Coffee Maker	Keurig	₹ 130.00	40	Kitchen	Coffee Maker Keurig
1-APR-US	11-APR	US	Smartphone	Samsung	₹ 900.00	25	Electronics	Smartphone Samsung
2-MAY-US	22-MAY	US	Backpack	North Face	₹ 70.00	20	Outdoor	Backpack North Face
7-JUN-UK	07-JUN	UK	Headphones	Sony	₹ 309.35	45	Electronics	Headphones Sony
9-JUL-UK	19-JUL	UK	T-Shirt	Adidas	₹ 30.00	5	Fashion	T-Shirt Adidas
3-AUG-UK	23-AUG	UK	Blender	Ninja	₹ 90.00	35	Kitchen	Blender Ninja
5-SEP-UK	05-SEP	UK	Tablet	Apple	₹ 500.00	50	Electronics	Tablet Apple
4-OCT-UK	14-OCT	UK	Hiking Boots	Timberland	₹ 130.00	10	Outdoor	Hiking Boots Timberland
7-JUN-IN	17-JUN	IN	Laptop	Hp	₹ 950.00	25	Electronics	Laptop Hp

- The Manufacturing Date information (date and month) was extracted from the Product ID using the LEFT function. The extracted data was then formatted into the DD-MM-YYYY format for better readability and consistency.

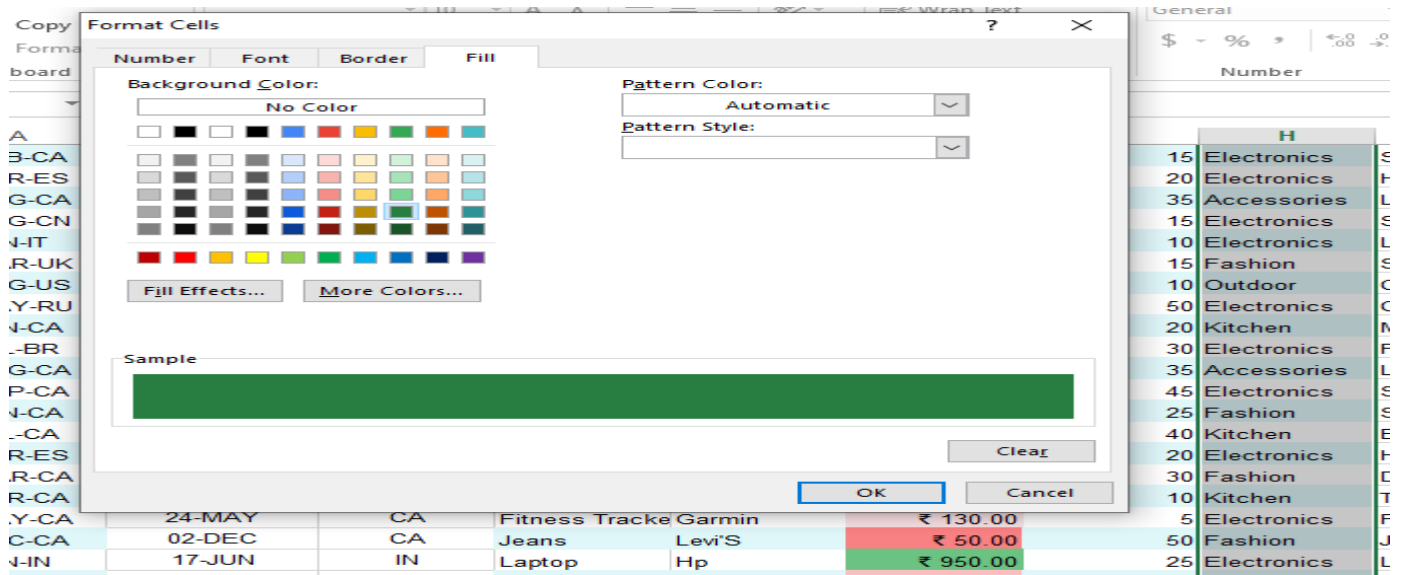
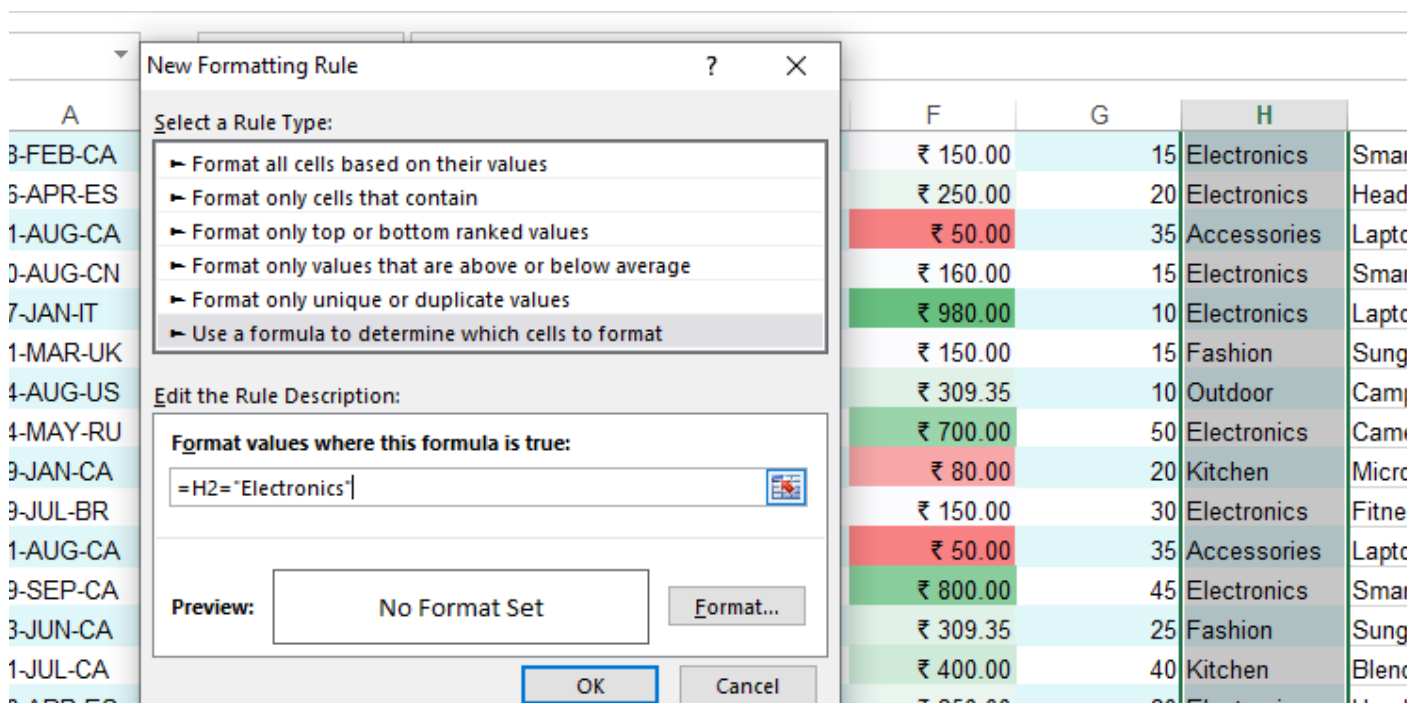
6. Conditional Formatting:

- Apply data bar or color scales conditional formatting in the "Price" column.

A	B	C	D	E	F	G	H
Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Manufacturing Date	Product Brand
28-JAN-US	Laptop	Dell	₹ 1,000.00	30	Electronics	28-JAN	Laptop Dell
15-FEB-US	Sneakers	Nike	₹ 80.00	15	Fashion	15-FEB	Sneakers Nike
03-MAR-US	Coffee Maker	Keurig	₹ 130.00	40	Kitchen	03-MAR	Coffee Maker Keurig
11-APR-US	Smartphone	Samsung	₹ 900.00	25	Electronics	11-APR	Smartphone Samsung
22-MAY-US	Backpack	North Face	₹ 70.00	20	Outdoor	22-MAY	Backpack North Face
07-JUN-UK	Headphones	Sony	₹ 309.35	45	Electronics	07-JUN	Headphones Sony
19-JUL-UK	T-Shirt	Adidas	₹ 30.00	5	Fashion	19-JUL	T-Shirt Adidas
23-AUG-UK	Blender	Ninja	₹ 90.00	35	Kitchen	23-AUG	Blender Ninja
05-SEP-UK	Tablet	Apple	₹ 500.00	50	Electronics	05-SEP	Tablet Apple
14-OCT-UK	Hiking Boots	Timberland	₹ 130.00	10	Outdoor	14-OCT	Hiking Boots Timberland
17-JUN-IN	Laptop	HP	₹ 950.00	25	Electronics	17-JUN	Laptop HP
25-NOV-AU	Sneakers	Adidas	₹ 90.00	40	Fashion	25-NOV	Sneakers Nike
08-DEC-DE	Coffee Maker	Nespresso	₹ 120.00	35	Kitchen	08-DEC	Coffee Maker Keurig
18-FEB-CA	Smartwatch	Fitbit	₹ 150.00	15	Electronics	18-FEB	Smartwatch Samsung
16-APR-ES	Headphones	Bose	₹ 250.00	20	Electronics	16-APR	Headphones Sony
21-AUG-CA	Laptop Bag	Samsonite	₹ 50.00	35	Accessories	21-AUG	Laptop Bag
20-AUG-CN	Smartwatch	Huawei	₹ 160.00	15	Electronics	20-AUG	Smartwatch

- Price column using Conditional Formatting. This provides a visual representation of price differences and helps quickly identify high and low-priced products.

2. Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."



Electronics								
A	B	C	D	E	F	G	H	I
Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Price (\$)	Quantity	Category	Product Brand
AN-US	28-JAN	US	Laptop	Dell	₹ 1,000.00	30	Electronics	Laptop Dell
EB-US	15-FEB	US	Sneakers	Nike	₹ 80.00	15	Fashion	Sneakers Nike
MAR-US	03-MAR	US	Coffee Maker	Keurig	₹ 130.00	40	Kitchen	Coffee Maker Keu
APR-US	11-APR	US	Smartphone	Samsung	₹ 900.00	25	Electronics	Smartphone Sams
MAY-US	22-MAY	US	Backpack	North Face	₹ 70.00	20	Outdoor	Backpack North F
UN-UK	07-JUN	UK	Headphones	Sony	₹ 309.35	45	Electronics	Headphones Sony
UL-UK	19-JUL	UK	T-Shirt	Adidas	₹ 30.00	5	Fashion	T-Shirt Adidas
UG-UK	23-AUG	UK	Blender	Ninja	₹ 90.00	35	Kitchen	Blender Ninja
SEP-UK	05-SEP	UK	Tablet	Apple	₹ 500.00	50	Electronics	Tablet Apple
OCT-UK	14-OCT	UK	Hiking Boots	Timberland	₹ 130.00	10	Outdoor	Hiking Boots Timb
UN-IN	17-JUN	IN	Laptop	Hp	₹ 950.00	25	Electronics	Laptop Hp
IOV-AU	25-NOV	AU	Sneakers	Adidas	₹ 90.00	40	Fashion	Sneakers Adidas
DEC-DE	08-DEC	DE	Coffee Maker	Nespresso	₹ 120.00	35	Kitchen	Coffee Maker Nes
EB-CA	18-FEB	CA	Smartwatch	Fitbit	₹ 150.00	15	Electronics	Smartwatch Fitbit
APR-ES	16-APR	ES	Headphones	Bose	₹ 250.00	20	Electronics	Headphones Bose
UG-CA	21-AUG	CA	Laptop Bag	Samsonite	₹ 50.00	35	Accessories	Laptop Bag Sams

Clipboard		Font		Alignment		Number	
▼	:	✕	✓	<i>f</i>	=TRIM(CLEAN(G2))		
A	B	C	D	E	F	G	H
Product ID		Brand Name	Price (\$)	Quantity		Category	Manufactur Date
28-JAN-US	Laptop	Dell	₹ 1,000.00	30	Electronics	Electronics	28-JAN
15-FEB-US	Sneakers	Nike	₹ 80.00	15		Fashion	15-FEB
03-MAR-US	Coffee Maker	Keurig	₹ 130.00	40		Kitchen	03-MAR
11-APR-US	Smartphone	Samsung	₹ 900.00	25		Electronics	11-APR
22-MAY-US	Backpack	North Face	₹ 70.00	20		Outdoor	22-MAY
07-JUN-UK	Headphones	Sony	₹ 309.35	45		Electronics	07-JUN
19-JUL-UK	T-Shirt	Adidas	₹ 30.00	5		Fashion	19-JUL
23-AUG-UK	Blender	Ninja	₹ 90.00	35		Kitchen	23-AUG
05-SEP-UK	Tablet	Apple	₹ 500.00	50		Electronics	05-SEP
14-OCT-UK	Hiking Boots	Timberland	₹ 130.00	10		Outdoor	14-OCT
17-JUN-IN	Laptop	HP	₹ 950.00	25		Electronics	17-JUN
25-NOV-AU	Sneakers	Adidas	₹ 90.00	40		Fashion	25-NOV
08-DEC-DE	Coffee Maker	Nespresso	₹ 120.00	35		Kitchen	08-DEC
18-FEB-CA	Smartwatch	Fitbit	₹ 150.00	15		Electronics	18-FEB

- Applied a custom Conditional Formatting rule to highlight "Electronics" entries in the Category column. This improves category visibility and data analysis.

